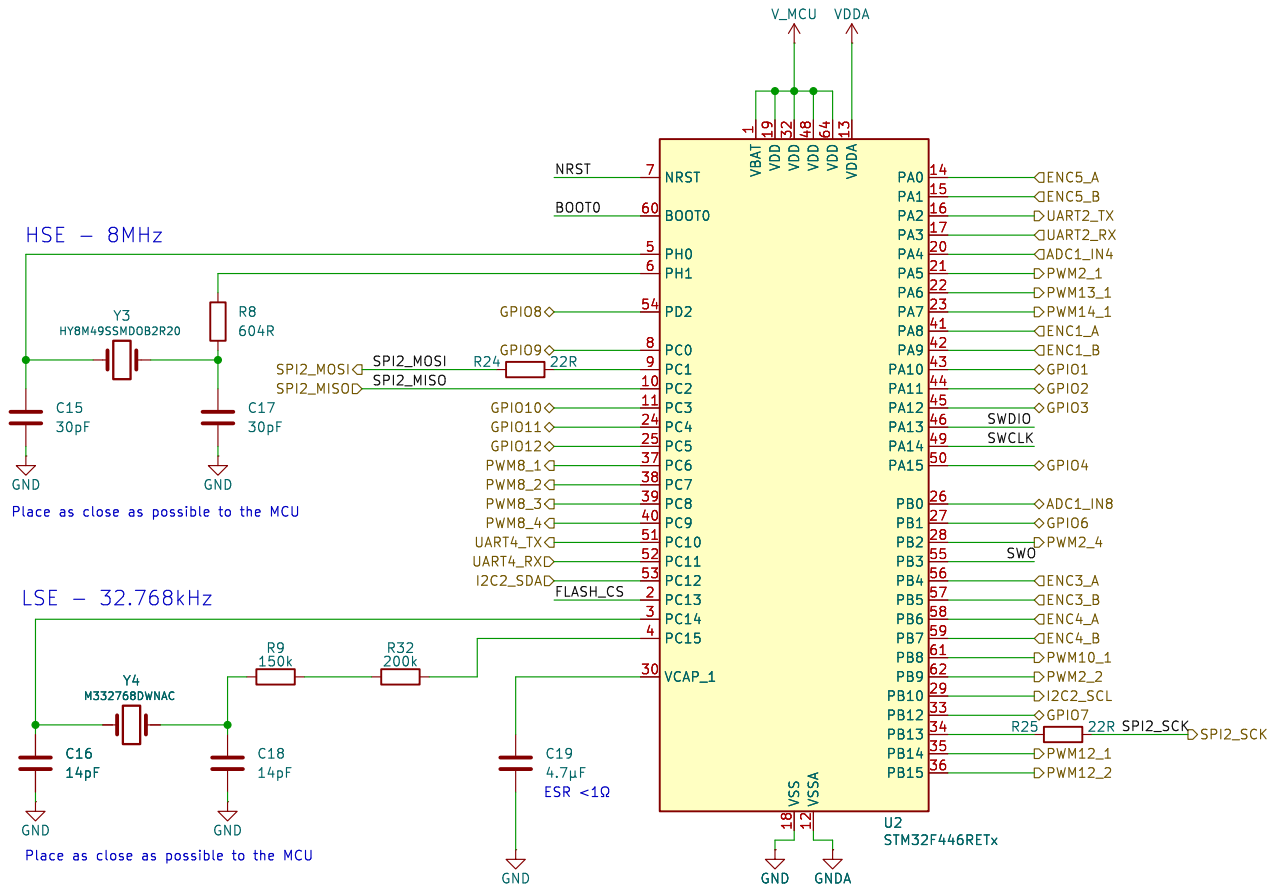
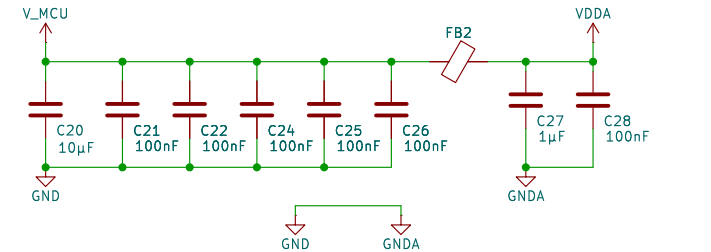




MCU



Power, Grounding & Decoupling

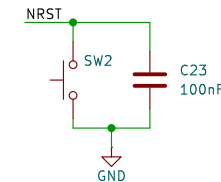


Decoupling strategy

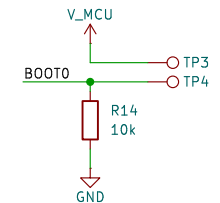
(Based on page 9 of AN4488)

- 10μF ceramic cap for the package.
- 100nF ceramic between pins 18-19, 31-32, 47-48 and 63-64.
- Vbat on pin 1 is connected to VDD with a 100nF decoupling cap.
- VDDA is connected to VDD through a ferrite bead.
- VDDA is decoupled with a 1μF and a 100nF cap between pins 12-13.
- GND is directly connected to the GND ground plane.

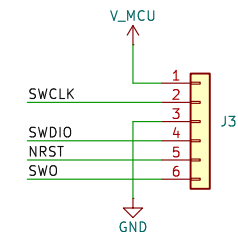
Reset



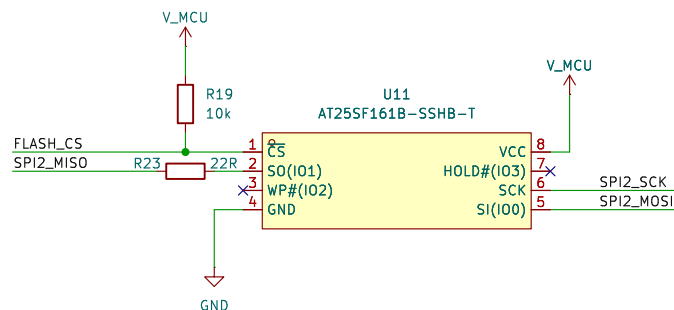
Boot



Debug



Flash (Config + Black Box)



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ElectronPlant

Sheet: /MCU/
File: mcu.kicad_sch

Title: MCU

Size: A4 Date: 2025-12-28

KiCad E.D.A. 9.0.6

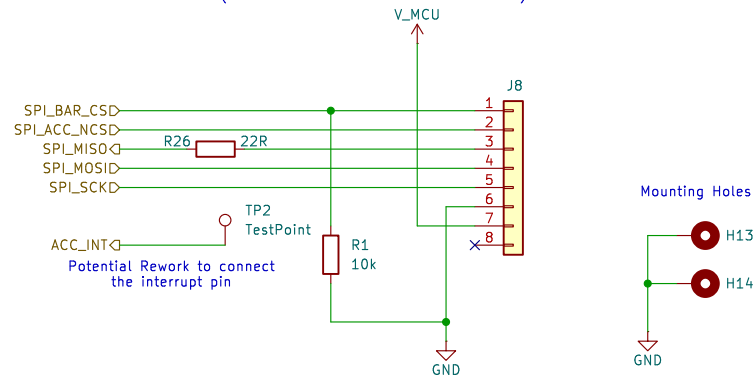
Rev: V0.0

Id: 2/10

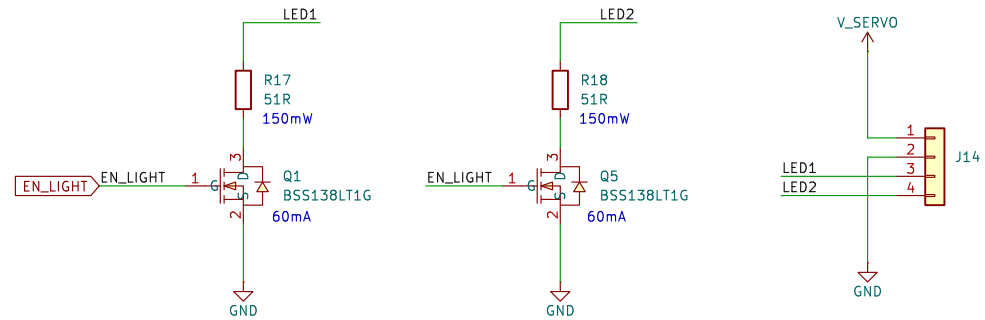
BeepyRC

Sensors

GY-91 (9-axis MPU9250 + BMP280) Module

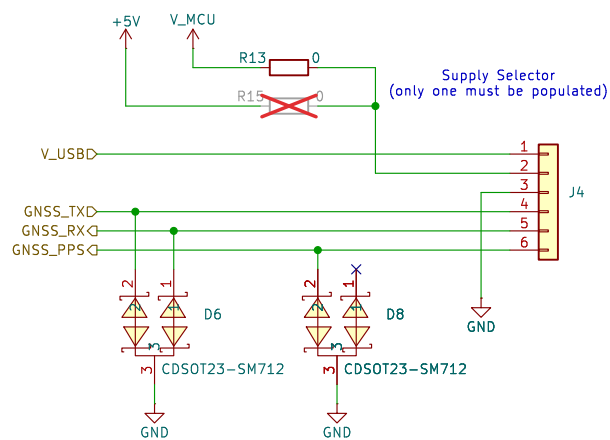


Lights

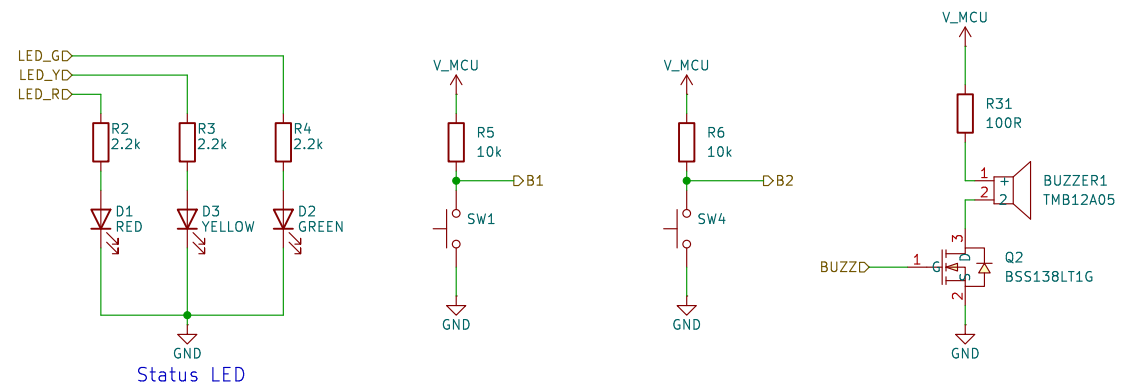


Current limiting resistor sizing:
 - LED forward voltage: ~3V (2.8 - 3.4)
 - Max forward current: 60mA

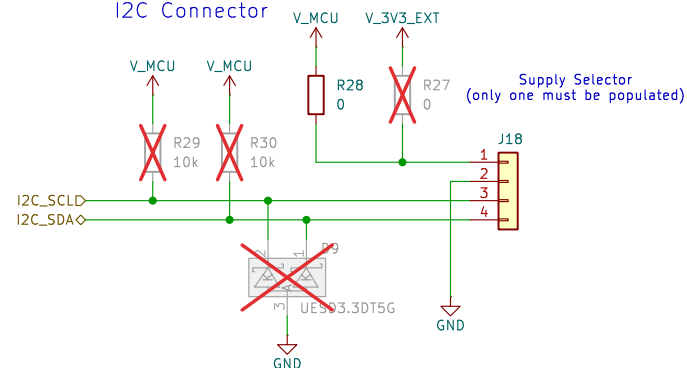
GNSS Module (L76K/VG7669T160N0MA)



User Interface



I2C Connector



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ElectronPlant

Sheet: /IO/

File: io.kicad_sch

Title: IO

Size: A4

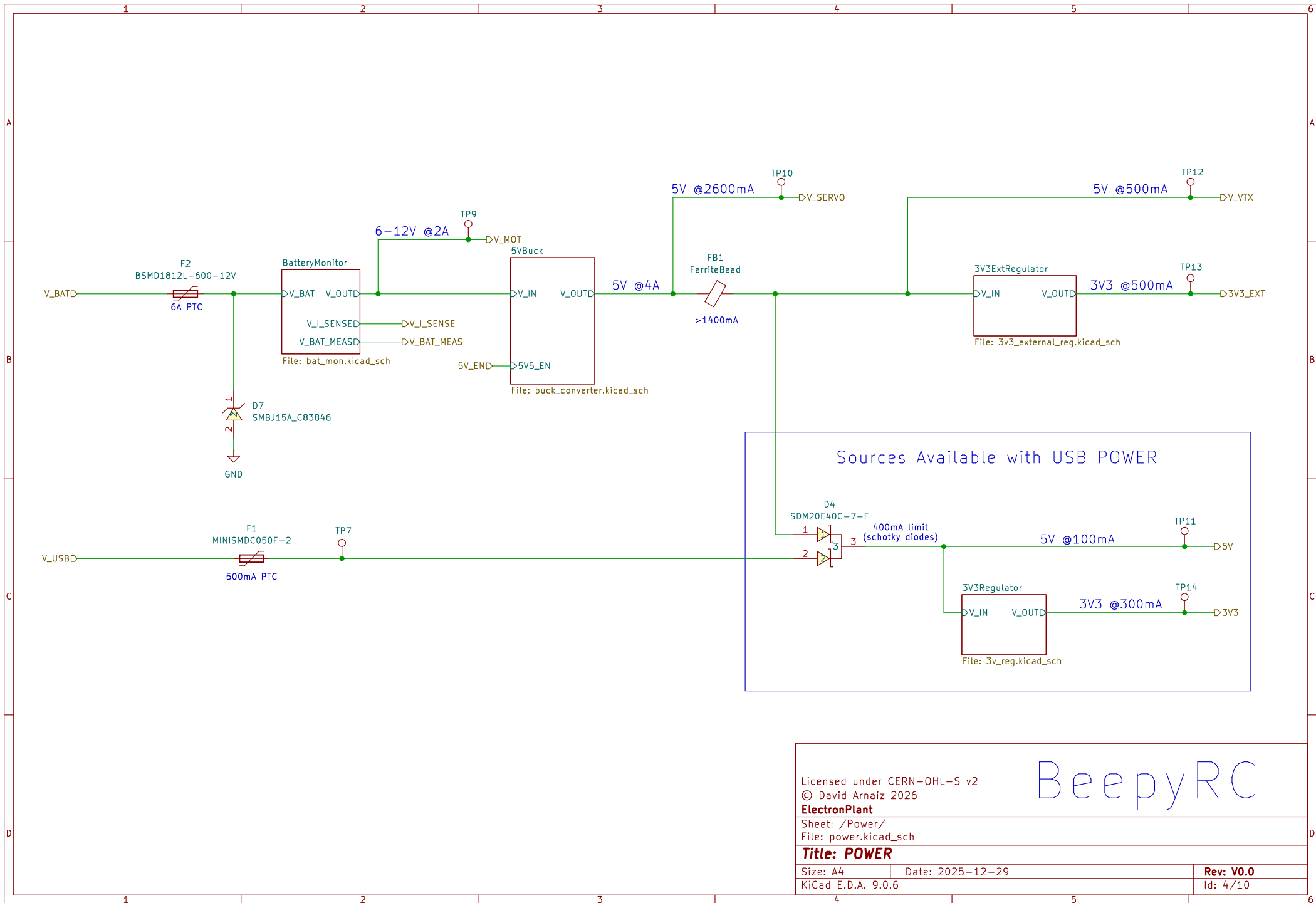
Date: 2025-12-29

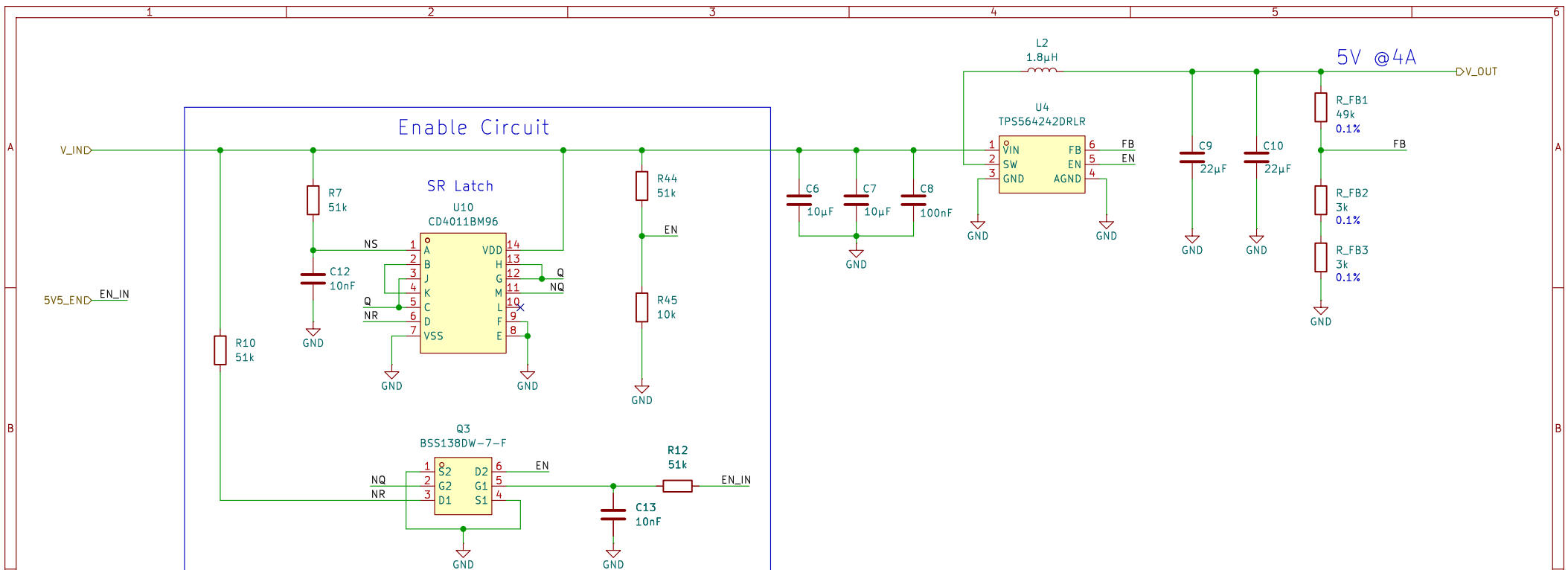
KiCad E.D.A. 9.0.6

Rev: V0.0

Id: 3/10

BeepyRC





Notes

--- Output capacitance ---

$$C_{OUT} = 44\mu F \text{ (} 22\mu F, 100\mu F \text{) [typ., (min, max)]}$$

$$I_{L_CAP} = (V_o * (V_{in} - V_o)) / (\sqrt{12} * V_{in} * L * f_{sw}) = 0.39A$$

L -> Inductor value
fsw -> Switching frequency (1200kHz)

--- Inductance ---

$$L (V_o = 5V) = 1.8 \text{ (} 1.5, 4.7 \text{) } \mu H \text{ [typ., (min, max)]}$$

Inductance Ratings:

$$I_{p-p} = (V_o / V_{in_max}) * (V_{in_max} - V_o) / (L * f_{sw}) = 1.35A$$

$$I_{peak} = I_o + I_{p-p} / 2 = 5.35A$$

$$I_{L_RMS} = \sqrt{I_o^2 + (1/12) I_{p-p}^2} = 4.3A$$

--- Feedback Resistors ---

$$V_o = 0.6 * (1 + R_{FB1} / R_{FB2})$$

$$R_{FB2} = (0.6 + 0.6 R_{FB1}) / V_o$$

--- EN input ---

When:

- EN is > 1.18(1.07, 1.33)V the module is enabled.
- EN is < 1(0.95, 1.2)V the module is disabled.

$$\text{Fixing } R_{EN1} + R_{EN2} \rightarrow R_{EN2} = (R_{EN1} + R_{EN2}) * V_{EN} / V_{IN}$$

$$V_{CUT} = V_{EN} * (R_{EN1} + R_{EN2}) / R_{EN2}$$

To limit to 6V input with $R_{EN1} + R_{EN2} = 50k$:

$$R_{EN2} \sim 8.2k$$

$$R_{EN1} \sim 40k$$

$$V_{CUT} = 5.87V \text{ (} 5.58, 7.05 \text{)}$$

This is too unreliable so it will be driven by the MCU!!

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Electron Plant

Sheet: /Power/5VBuck/

File: buck_converter.kicad_sch

Title: Buck Converter

Size: A4

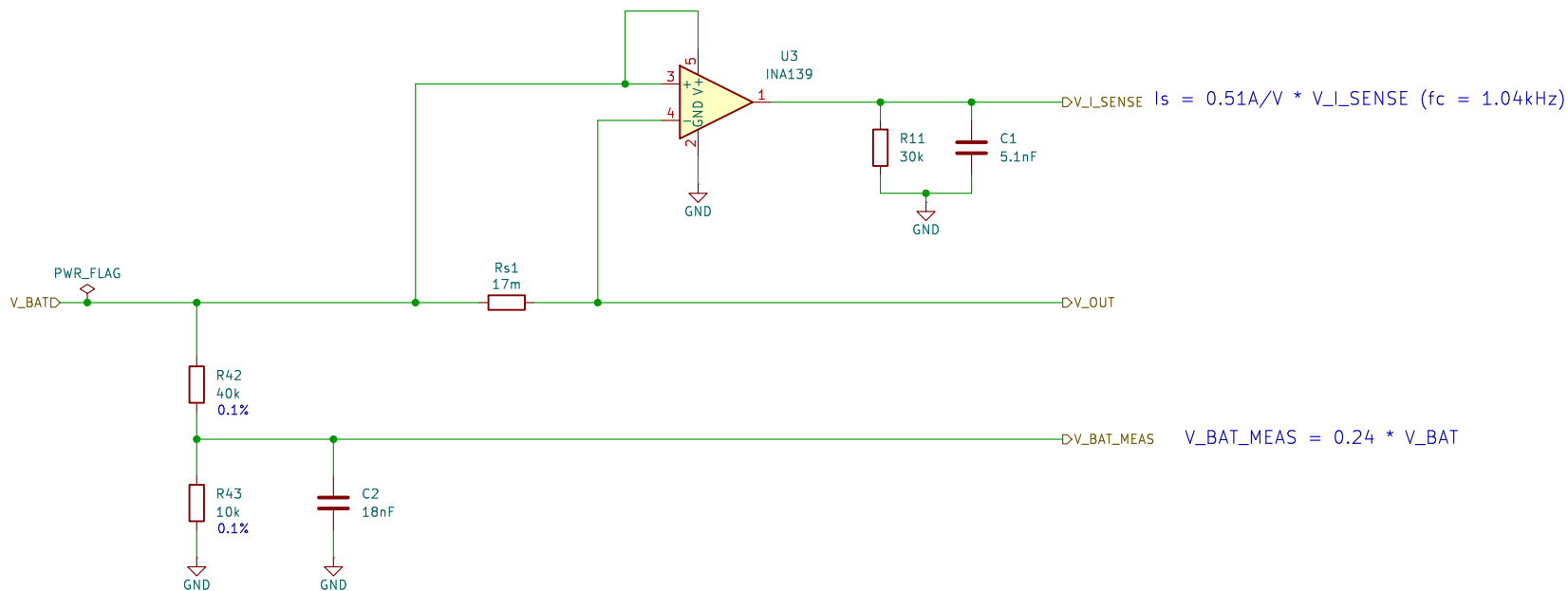
Date: 2026-02-08

KiCad E.D.A. 9.0.6

Rev: V0.0

Id: 5/10

BeepyRC



--- Battery Voltage ---

2S battery range 6 – 8.4V, with 7.4 nominal
Maximum ranges 6 – 12V --> Limited by the N20 motors
R_SENSE is set to:

1. Be <50kOhm (ADC external resistance limit),
2. V_BAT_MEAS is ~3V @ 12V
3. Low-pass filter with Fc ~1kHz

--- INA139 ---

Output Voltage: $V_o = I_s * 1mA/V * R_L$
Vs full scale = 100mV (could also be 50mV to 500mV)
R_L -> Set to generate ~3V at 6A (<50kOhm, ADC external resistance limit)
C_L -> Low pass filter with a cut-off frequency at ~1kHz

--- Shunt Resistor ---

Vs = 100mV with max current
 $\max(I_s) = 6A > (4 * 0.35A \text{ for the motors} + 4A \text{ for the 5.5V rail})$
 $R_s = V_s / \max(I_s) = \sim 17mOhm$
Power Rating: 0.6W

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Electron Plant

Sheet: /Power/BatteryMonitor/

File: bat_mon.kicad_sch

Title: Battery Monitor

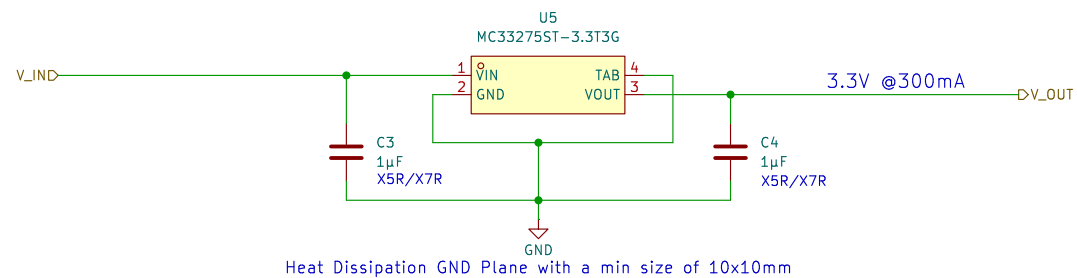
Size: A4 Date: 2026-02-11

KiCad E.D.A. 9.0.6

Rev: V0.0

Id: 6/10

BeepyRC



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ElectronPlant

Sheet: /Power/3V3Regulator/

File: 3v_reg.kicad_sch

Title: 3V3 Regulator

Size: A4

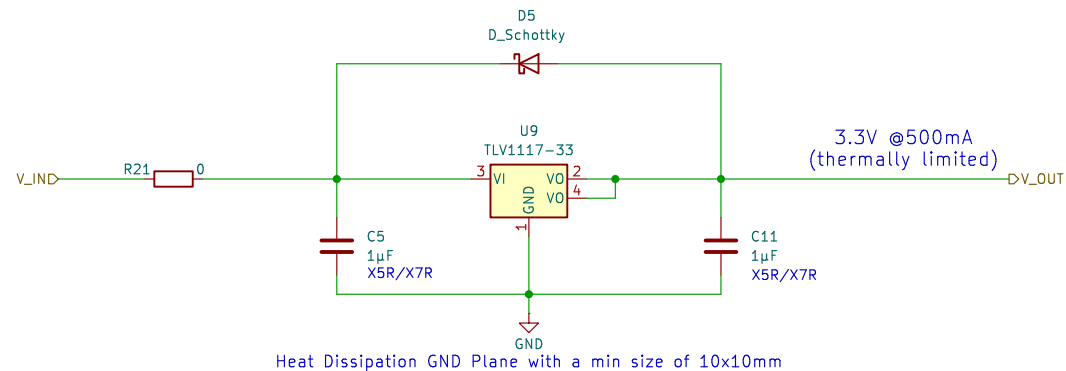
Date: 2026-02-25

KiCad E.D.A. 9.0.6

Rev: V0.0

Id: 8/10

BeepyRC



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Sheet: /Power/3V3ExtRegulator/
File: 3v3_external_reg.kicad_sch

Title: 3V3 External Regulator

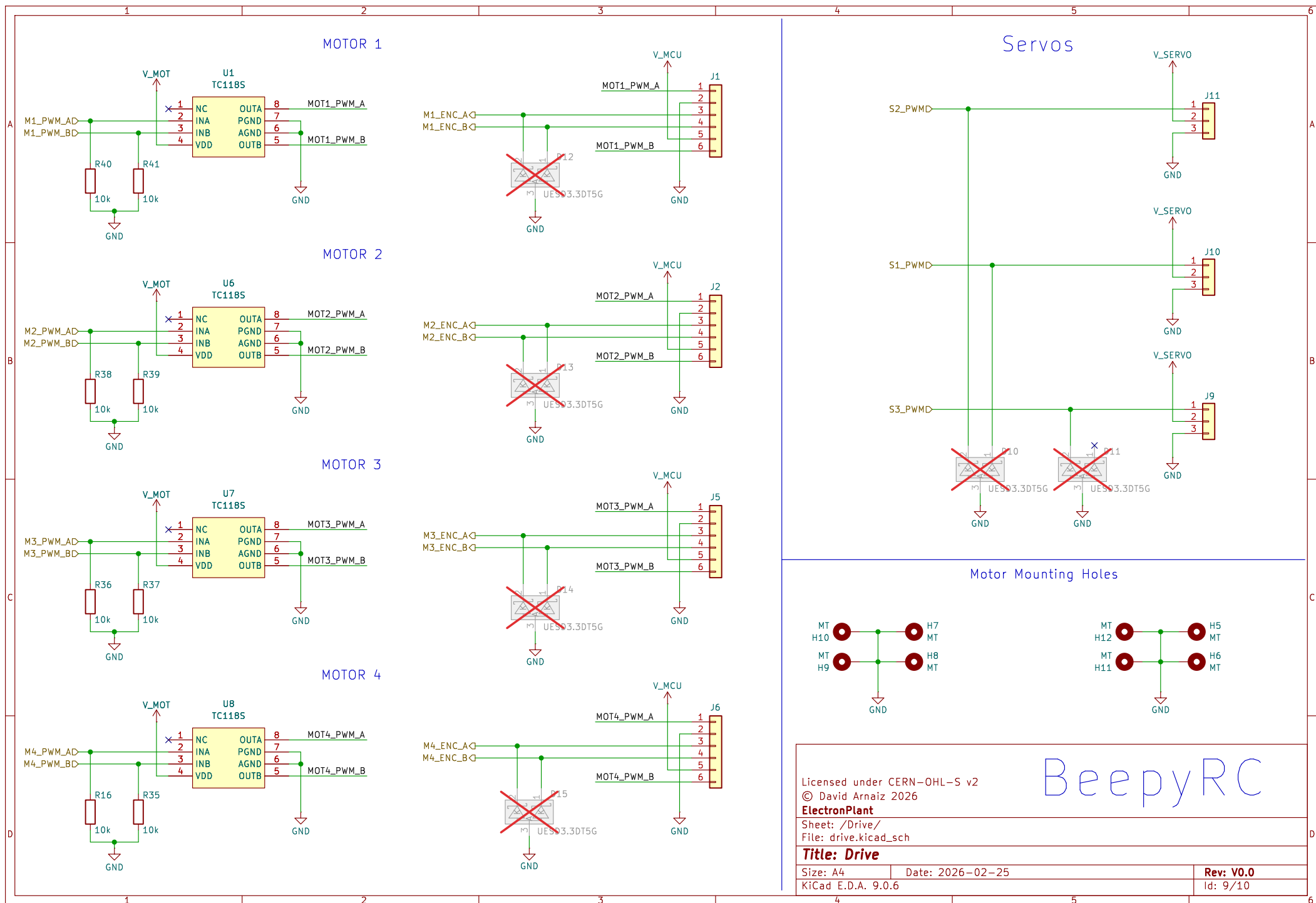
Size: A4 Date: 2026-02-25

KiCad E.D.A. 9.0.6

Rev: V0.0

Id: 8/10

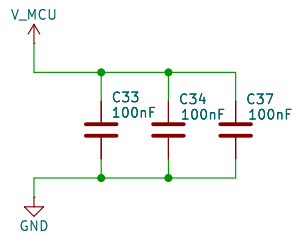
BeepyRC



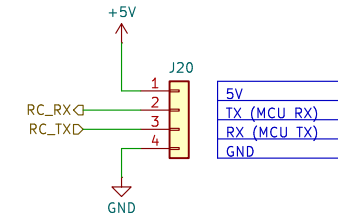
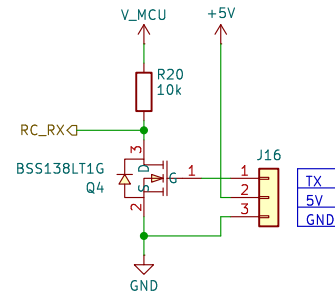
OSD

Decoupling strategy

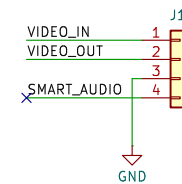
C33 between AVDD (P12) and AGND (P10)
C34 between DVDD (P2) and DGND (P1)
C37 between PVDD (P14) and PGND (P11)



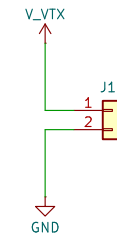
RC Serial



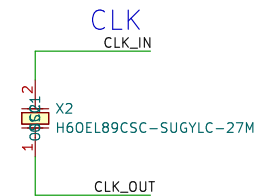
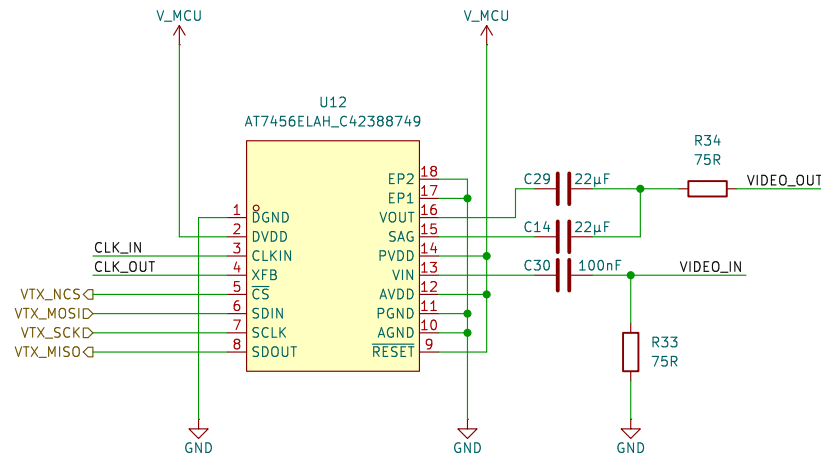
VTX



VTX Connector



Power Connector



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ElectronPlant

Sheet: /RC/

File: rc.kicad_sch

Title: RC

Size: A4

Date: 2026-02-25

KiCad E.D.A. 9.0.6

Rev: V0.0

Id: 10/10

BeepyRC